Occupational Zoonotic Transmission of Virulent Newcastle Disease Virus in Nigeria, 2023

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Abstract

A virulent Newcastle disease virus (NDV) sub-genotype XIV.2 outbreak in a fully vaccinated broiler parent flock in Maiduguri, Nigeria resulted in 95% mortality. During field investigation, the attending veterinarian developed severe conjunctivitis and mild systemic symptoms following unprotected exposure. Molecular characterization revealed a polybasic F protein cleavage site (^{112}RRQKR↓F^{117}) and amino acid substitutions (R114Q, V118I, A220V) within predicted T-cell epitopes, suggesting immune evasion from genotype-matched vaccines. This outbreak underscores the occupational zoonotic risk of velogenic NDV and highlights the need for genotype-matched vaccination, enhanced biosafety protocols, and integrated One Health surveillance.

Background

- Newcastle disease (ND) remains a major threat to poultry production in Nigeria, where sub-genotype XIV.2 dominates despite routine vaccination.
- NDV is zoonotic: human cases usually present with conjunctivitis, though severe pneumonia can occur in immunocompromised individuals.
- In November 2023, a fully vaccinated broiler flock in Maiduguri suffered 95% mortality, and the attending veterinarian developed conjunctivitis after unprotected exposure.
- Repeated vaccine failures suggest genotype–vaccine mismatch and potential immune-evasion mutations in XIV.2 strains.

Objectives

- Molecularly characterize XIV.2 outbreak strain
- Identify T-cell epitopes and assess immune-evasion signatures.
- Simulate vaccine-induced immune responses against XIV.2

Materials & Methods

- **Virus Isolation & Sequencing**
  - Virus isolated in 10-day-old embryonated eggs; Complete F gene sequenced
  - Phylogenetic analysis: RAxML (1000 bootstrap)
- **Immunoinformatics**
  - Epitope prediction: MHC class I/II (IEDB server)
  - Mutation analysis: XIV.2 vs. vaccine strains
  - MHC modeling and Molecular docking: AlphaFold3 & AutoDock Vina
  - Immune simulation: C-ImmSim
    - Scenario 1: LaSota vaccine → LaSota virus challenge
    - Scenario 2: LaSota vaccine → XIV.2 virus challenge



Figure 4. Conjunctivitis observed in the exposed veterinarian following direct contact with infected birds. **(A)** Unilateral conjunctivitis. **(B)** Bilateral conjunctivitis. Written informed consent for the use and publication of the images was obtained from the affected individual.

Results

- **Occupational infection:** Veterinarian developed conjunctivitis, fever, and fatigue following unprotected exposure.
- **Molecular identification:** Outbreak strain clustered within NDV sub-genotype XIV.2/XIVb, sharing 98% identity with circulating Nigerian strains.
- **Virulence markers:** F protein contained polybasic cleavage motif (^{112}RRRKR↓F^{117}), consistent with high mortality.
- **Epitope mutations:** Key substitutions (R114Q, V118I, A220V) detected within predicted MHC-I/II epitopes, diverging from LaSota, Komarov, Hitchner-B1, Mukteswar vaccine sequences.
- **Vaccine mismatch:** Simulated LaSota–Komarov prime–boost induced humoral response but failed to clear XIV.2 challenge in silico.

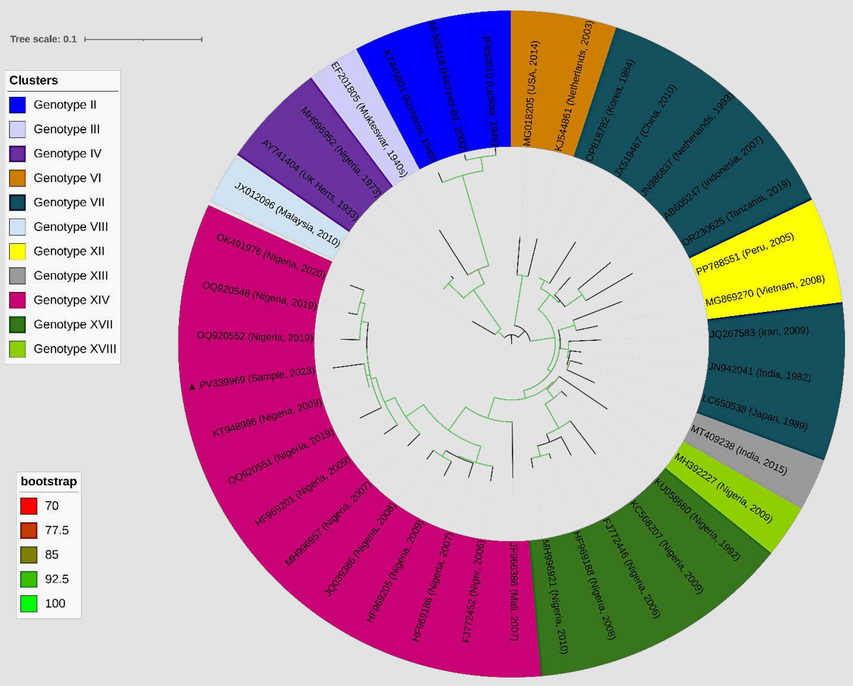


Figure 1. ML phylogenetic tree of NDV full F gene sequences. The outbreak sample (▲PV339969) clusters within genotype XIV (pink) and shows clear divergence from commonly used vaccine strains (blue).

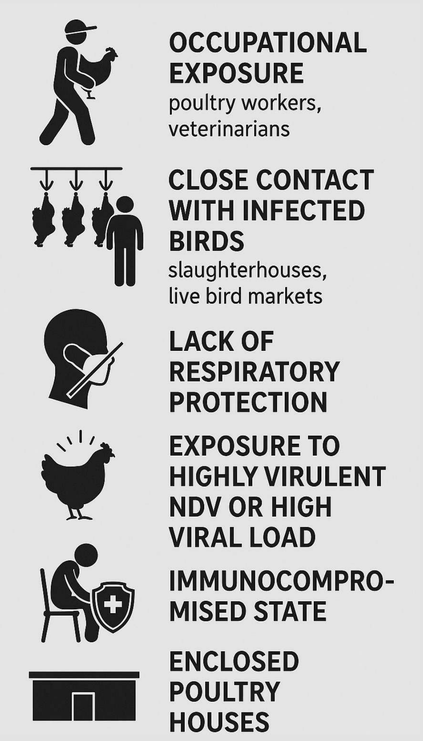


Figure 2. Illustration summarizing key risk factors associated with zoonotic transmission of Newcastle disease virus.

Table 1. (A) Mutation comparison of XIV.2 T-cell epitope peptides with vaccine strains (genotype II) and genotypes VII, XVII, and XVIII. Amino-acid differences highlight genotype-specific divergence with implications for immune recognition.

Peptides (XIV.2)	Peptide 1	Peptide 2	Peptide 3	Peptide 4	Peptide 5	Peptide 6
	RRKRFVGA	TAFGPQJTS	YTDPYPLVF	MAKSSYKPR	RAKFAWLHI	SVAVSYSLK
LaSota	RQGRLIGA	TVFGPQJTS	YTDPYPLIF	MAKSSYKGR	TAKFAWLHI	NVAVSYSLK
Hitchner-B1	RQGRLIGA	TVFGPQJTS	YTDPYPLIF	MAKSSYKGR	TAKFAWLHI	NVAVSYSLK
Komarov	RQKRFIGA	TVFGPQJTS	YTDPYPLIF	MAKSSYKGR	TAKFAWLHI	DVAVSYSLK
Mukteswar	RQRRFIGA	TVFGPQJTS	YTDPYPLVF	MAKSSYKGR	TAKFAWLHI	NVAVSYSLK
VII.2	RRKRFIGA	TVFGPQJTS	YTDPYPLIF	MAKSSYKPQ	RAKFAWLHI	SVAVSYSLK
XVII	RQKRFIGA	TVFGPQJTS	YTDPYPLVF	MAKSSYKPR	RAKFAWLHI	SVAVSYSLK
XVIII	RQKRFIGA	TVFGPQJTS	YTDPYPLVF	MAKSSYKPR	RAKFAWLHI	SVAVSYSLK

Table 2. Molecular docking scores of LaSota and XIV.2 peptides with chicken MHC I. Lower affinity of XIV.2 indicates potential immune evasion.

Strain/ Control	Epitope	Receptor	Affinity (kcal/mol)
XIV.2	RRKRFVGA	Chicken MHC class I (BF2*2101)	-6.3
	TAFGPQJTS		-6.9
	MAKSSYKPR		-5.9
	SVAVSYSLK		-7.9
LaSota (Genotype II)	RQGRLIGA		-6.7
	TVFGPQJTS		-7.3
	MAKSSYKGR		-7.5
	NVAVSYSLK		-7.7
Positive Control	IDWFDGKE		-7.0
	RRALREGY		-6.9

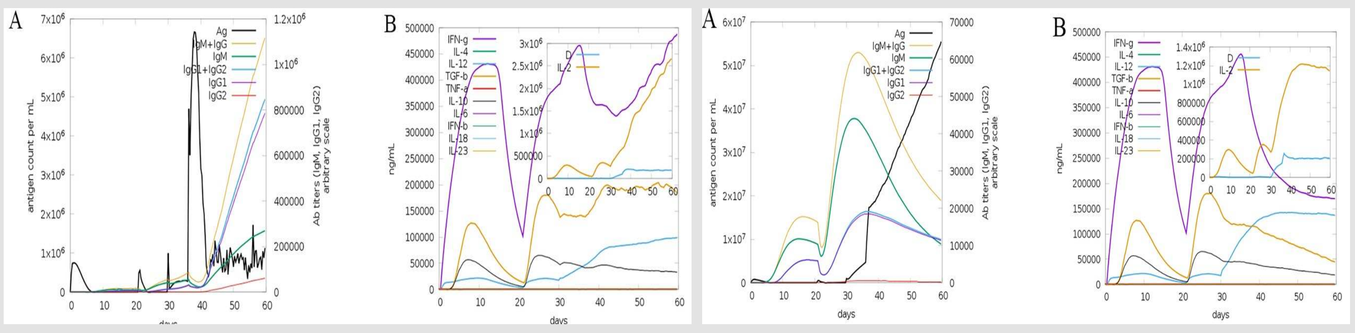


Figure 3. (Left) Immune response to LaSota–Komarov vaccination followed by homologous LaSota challenge. **(Right)** Response to the same vaccination followed by heterologous XIV.2 challenge. Homologous LaSota challenge shows strong memory activation, whereas XIV.2 induces a delayed and weaker response, indicating reduced cross-protection and potential genotype-specific immune evastion.

Conclusion

- NDV sub-genotype XIV.2 exhibited clear **vaccine mismatch**, showing reduced cross-protection after LaSota–Komorov vaccination.
- A zoonotic conjunctivitis case in the attending vet reinforces **NDV's zoonotic potential** and the elevated infection risk among poultry workers and veterinarians.
- These findings emphasize the need for **genotype-matched vaccines** and **stronger biosafety measures** across the wildlife–poultry–human interface.

References

1) Sajo MU, et al. Field outbreak investigation and immunoinformatic analysis suggest potential immune evasion by Newcastle disease virus Sub-Genotype XIV.2 in Nigeria. *Sci Rep.* 2025.

2) Ul-Rahman A, et al. Zoonotic potential of Newcastle disease virus: Old and novel perspectives related to public health. *Rev Med Virol*. 2022.